

# HEART DISEASE PREDICTION USING MACHINE LEARNING

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## Project Title

### Heart Disease Prediction Using Machine Learning

## Objective

The objective of this project is to develop a machine learning model that predicts the likelihood of heart disease based on a patient's clinical and medical attributes. The model aims to assist healthcare professionals in early diagnosis, enabling timely treatment and improving patient outcomes.

## Dataset Overview

The Heart Disease dataset consists of patient health records used to predict the presence of heart disease. It is a binary classification problem where:

- **1** – Presence of Heart Disease
- **0** – No Heart Disease

The dataset includes features such as:

- Age
- Sex
- Chest Pain Type
- Resting Blood Pressure
- Cholesterol
- Fasting Blood Sugar
- Resting ECG
- Maximum Heart Rate Achieved
- Exercise-Induced Angina
- ST Depression (Old peak)

- Slope of Peak Exercise ST Segment
- Number of Major Vessels
- Thalassemia

## Industry Name

This project belongs to:

- Healthcare Analytics

## Data Preprocessing & Feature Engineering

### 1. Data Cleaning

- Loaded the dataset into Python.
- Checked for missing values and duplicate records.
- Verified data quality before model training.

### 2. Exploratory Analysis

- Examined data distributions and feature statistics.
- Identified relationships between features and the target variable

### 3. Data Preparation

- Split the dataset into training and testing sets.
- Applied preprocessing techniques where required for machine learning models.

## Exploratory Data Analysis (EDA)

EDA was performed to understand the dataset before training the models.

Key observations include:

- The target variable distribution was analyzed.
- Correlation analysis identified relationships among features.
- Histograms and count plots were used to visualize feature distributions.
- A heatmap was created to study feature correlations.
- These analyses helped identify the most influential variables for prediction

## Model Building

The following machine learning algorithms were trained and evaluated to predict whether a patient has heart disease.

### Logistic Regression

- Logistic Regression was used as a baseline classification model. It predicts the probability of a patient having heart disease based on the input features.

### Decision Tree

- Decision Tree classifies patients by splitting the data into different conditions based on important features. It is easy to understand and helps identify the factors influencing the prediction.

### Random Forest

- Random Forest combines multiple decision trees to make more accurate and reliable predictions. It reduces overfitting and provided the best overall performance in this project.

### K-Nearest Neighbors (KNN)

- KNN predicts the class of a patient by comparing it with similar patients in the dataset. It classifies a new patient based on the majority class of its nearest neighbors.

### Support Vector Machine (SVM)

- SVM classifies patients by finding the optimal boundary that separates those with heart disease from those without it. It performs well on classification problems with clear class separation.
- Each model was trained using the training dataset and evaluated on the testing dataset. Their performance was compared using accuracy and other evaluation metrics to identify the best model.

## Model Evaluation

The performance of all machine learning models was evaluated using the following metrics:

- **Accuracy Score:** Measures the percentage of correct predictions made by the model.
- **Confusion Matrix:** Shows the number of correct and incorrect predictions for each class, helping to understand the model's performance in detail.
- **Classification Report:** Provides important evaluation metrics such as precision, recall, F1-score, and accuracy to assess the effectiveness of each model.

After comparing the performance of all the models, **Random Forest** achieved the best overall results with the highest accuracy and more reliable predictions. Therefore, it was selected as the most suitable model for heart disease prediction.

| Machine Learning Model       | Accuracy |
|------------------------------|----------|
| Logistic Regression          | 83.33%   |
| Decision Tree                | 77.78%   |
| Random Forest                | 86.11%   |
| K-Nearest Neighbours (KNN)   | 83.33%   |
| Support Vector Machine (SVM) | 83.33%   |

## Key Observations

- **Random Forest** achieved the highest accuracy of **86.11%**, making it the best-performing model.
- **Logistic Regression**, **KNN**, and **SVM** each achieved an accuracy of **83.33%**, providing consistent and reliable results
- **Decision Tree** achieved an accuracy of **77.78%** and showed signs of overfitting compared to the other models.
- Comparing multiple machine learning algorithms helped identify the most suitable model for predicting heart disease.
- The project demonstrates that machine learning can effectively support the early prediction of heart disease using patient health data.

## Business Interpretation

The machine learning model helps identify patients who may be at risk of heart disease based on their medical data. Early prediction enables timely diagnosis and treatment, improves patient care, and supports healthcare professionals in making informed decisions. This model can serve as a useful decision-support tool for hospitals and clinics.

## Final Conclusion

The Heart Disease Prediction project was successfully completed by developing and evaluating multiple machine learning models using patient health data. The models were compared based on their performance using evaluation metrics such as accuracy, confusion matrix, and classification report. Among all the models, **Random Forest** achieved the best overall performance due to its high accuracy and reliable predictions.

This project demonstrates that machine learning can effectively support the early prediction of heart disease, helping healthcare professionals make informed decisions and enabling timely diagnosis and treatment. With further improvements and larger datasets, the model can be enhanced for practical healthcare applications.

## **Future Improvements**

- Perform hyperparameter tuning to further improve model performance.
- Train the model using larger and more diverse datasets.
- Deploy the model as a web application using Flask.
- Integrate the model into healthcare decision support systems

## **Tech Stack**

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-learn

## **Challenges Faced During the Project**

### **1. Understanding the Dataset**

The dataset contained multiple medical features, requiring careful analysis to understand how each feature influenced the prediction of heart disease. Exploratory Data Analysis (EDA) was performed to identify important patterns and relationships.

## 2. Data Preprocessing

Preparing the dataset involved checking for missing values, removing duplicate records, and ensuring the data was clean and ready for model training. Proper preprocessing helped improve the quality of the predictions.

## 3. Model Selection

Different machine learning algorithms produced different levels of accuracy and performance. Multiple models were trained and compared to identify the most suitable algorithm for heart disease prediction.

## 4. Model Evaluation

Interpreting evaluation metrics such as accuracy, precision, recall, F1-score, and the confusion matrix was important for comparing the models and selecting the best-performing one.

## 5. Overfitting

The Decision Tree model showed signs of overfitting, where it performed well on the training data but less effectively on unseen test data. Random Forest helped overcome this issue by providing more stable and reliable predictions.

## 6. Selecting the Best Model

After evaluating all the models, Random Forest was selected as the final model because it achieved the best overall performance, provided consistent predictions, and generalized well to new data.

## References

1. Heart Disease Dataset ( <https://d3ilbtxij3aepc.cloudfront.net/projects/CDS-Capstone-Projects/PRCP-1016-HeartDiseasePred.zip>).
2. Scikit-learn Documentation.
3. Pandas Documentation.
4. NumPy Documentation
5. Matplotlib Documentation.
6. Seaborn Documentation.



## Acknowledgement

his project was developed using the **Heart Disease Prediction** dataset provided through **Datamites** as part of the Artificial Intelligence and Machine Learning program. The project leverages open-source Python libraries including **Pandas, NumPy, Scikit-learn, Matplotlib, and Seaborn** for data preprocessing, visualization, machine learning model development, and predictive analysis.